

A short mathematical analysis of the wash-bay cleaning simulation

floor_clean

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Abstract

This note extracts the closed-form scaling laws buried in the simulator and shows that the project’s headline empirical finding—*coverage beats technique*—is not an accident of the experiments but a direct consequence of two numbers: the jet removes about 845 g/m² of grit per pass from a floor carrying 20 g/m². With a removal capacity 40× the load, the cleaning rate saturates and the only free variable left is swept area. Everything else in the model—the standoff cliff, the angle sensitivity, the uselessness of extra water for cutting—follows from the impingement pressure law $P \propto \cos^4 \theta d^{-2}$ and the Manning normal-depth law $h \propto Q^{3/5}$.

1 The impingement pressure law

The wand tip sits at height d (standoff) above the slab, tilted θ from vertical. The beam travels a slant distance $L = d/\cos \theta$ and lands $d \tan \theta$ ahead of the operator. A fan tip of full angle ϕ spreads linearly in L ; the sheet thickness starts at t_0 and grows at a full angle ψ , and oblique incidence stretches it by $1/\cos \theta$:

$$w(L) = 2L \tan \frac{\phi}{2}, \quad t(L, \theta) = \frac{t_0 + 2L \tan \frac{\psi}{2}}{\cos \theta}. \quad (1)$$

The footprint is modelled as a Gaussian with $\sigma_u = w/4$, $\sigma_v = t/4$, so its effective area is $A = 2\pi\sigma_u\sigma_v = \pi wt/8$. Air drag bleeds the orifice velocity V_0 over an e -folding length Λ , and only the normal component of the momentum flux presses on the slab:

$$P(d, \theta) = \frac{\rho Q V_0 e^{-L/\Lambda} \cos \theta}{A(L, \theta)} = \frac{8 \rho Q V_0 e^{-L/\Lambda} \cos^2 \theta}{2\pi L \tan \frac{\phi}{2} (t_0 + 2L \tan \frac{\psi}{2})} \quad (2)$$

In the far field ($2L \tan \frac{\psi}{2} \gg t_0$, i.e. beyond $L \approx 9$ cm) the two linear spreads combine into an inverse-square law and the $\cos \theta$ factors collect:

$$P \sim \frac{2\rho Q V_0}{\pi \tan \frac{\phi}{2} \tan \frac{\psi}{2}} \cdot \frac{\cos^4 \theta}{d^2} e^{-d/(\Lambda \cos \theta)}. \quad (3)$$

Equation (3) is the single most consequential line in the model. The fourth power of $\cos \theta$ is the reason wand angle dominates: two of the powers are the oblique projection of the momentum, and two come from the footprint being stretched in *both* directions by the oblique strike. With the shipped parameters (1200 psi, 4 gpm, $C_d = 0.95$, so $V_0 = 122$ m/s and a thrust of $\rho Q V_0 = 30.8$ N):

technique	d	θ	lead	P	P/Y_0
vertical, 12 in	0.305 m	0°	0 in	35.2 kPa	14.1
far_to_near baseline	0.400 m	31.5°	9.7 in	10.6 kPa	4.23
operator, 12 in lead	0.305 m	45°	12 in	8.70 kPa	3.48
operator, 18 in lead	0.305 m	56°	17.8 in	3.28 kPa	1.31
sweep phase	0.300 m	72°	36 in	0.26 kPa	0.10

against a base adhesion $Y_0 = 2.5 \text{ kPa}$ and a worn-lane centre $Y_w = 5.5 \text{ kPa}$. A 11° wrist movement—the whole span from a 12 in lead to an 18 in lead—costs a factor $(\cos 45^\circ / \cos 56^\circ)^4 = 2.65$ in cutting pressure and carries the jet from $1.58 Y_w$ (cuts the worn lanes) to $0.60 Y_w$ (does not). That the operator’s own stated working range straddles a threshold is a prediction of (3), not a fitted observation.

2 Why the threshold is soft: the patch integral

The impact patch is thinner than one grid cell (2.3 cm against 7 cm), so per-cell sampling of the Gaussian would be a grid-alignment lottery. The simulator instead integrates the excess pressure analytically over the patch. For a Gaussian of peak P above a yield stress Y , writing s for the squared Mahalanobis radius,

$$\int \max(0, P e^{-s/2} - Y) dA = A_{\text{patch}} \underbrace{\left(P - Y - Y \ln \frac{P}{Y} \right)}_{G(P, Y)}, \quad (4)$$

the region of integration terminating at $s_{\text{max}} = 2 \ln(P/Y)$. Near threshold, with $\varepsilon = P/Y - 1$,

$$G = Y(\varepsilon - \ln(1 + \varepsilon)) = \frac{1}{2} Y \varepsilon^2 + O(\varepsilon^3), \quad (5)$$

so cutting switches on *quadratically*, not as a step—and $G \rightarrow P$ when $P \gg Y$. Two corollaries matter. First, this is a C^1 threshold, which is why the physics is differentiable through the cutting onset. Second, marginal technique is punished twice: at $P/Y = 1.58$ the excess G is only 306 Pa, 7% of the 4462 Pa the baseline delivers, even though the pressure ratio differs by just $2.7\times$.

The same integral gives the *effective swath*—the width of floor actually cut on a pass—as the chord where the profile still exceeds Y :

$$W_{\text{eff}}(d, \theta) = 2\sigma_u \sqrt{2 \ln(P/Y)} = L \tan \frac{\phi}{2} \sqrt{2 \ln(P/Y)}. \quad (6)$$

This is non-monotonic in d : the fan widens linearly while the pressure falls as d^{-2} , so W_{eff} peaks near 0.4 m and then collapses.

d (m)	0.20	0.30	0.40	0.60	0.70	0.80
P (kPa)	42.4	19.2	10.6	4.28	2.97	2.14
$W_{\text{eff, base}}$ (cm)	12.4	15.8	17.7	16.2	10.6	0.0
$W_{\text{eff, worn lane}}$ (cm)	10.5	12.3	11.9	0.0	0.0	0.0

The cliff at $d^* \approx 0.80 \text{ m}$ (31 in) reported in `ANSWER.md` is exactly the root of $P(d, 31.5^\circ) = Y_0$, and the worn lanes lose their swath at 0.54 m (21 in) — 26 cm lower. Because P is proportional to momentum flux and independent of ambient film depth, d^* is water-invariant, which is what the flow sweep confirmed.

3 The main result: the job is coverage-limited by a factor of 40

The removal rate of adhered grit is $\dot{m} = k G(P, Y)$ with $k = 2.5e - 3 \text{ kg/m}^2/\text{s/Pa}$. A point on the floor is under the jet only for the time it takes the patch thickness t to sweep past at walking speed v :

$$\tau_{\text{dwell}} = \frac{t(L, \theta)}{v} = \frac{22.7 \text{ mm}}{0.3 \text{ m/s}} = 76 \text{ ms}. \quad (7)$$

So the grit a single pass is *capable* of lifting is

$$m_{\text{cap}} = k G \tau_{\text{dwell}} = (2.5 \times 10^{-3}) \times 4462 \times 0.076 = 845 \text{ g/m}^2, \quad (8)$$

against a floor loading $m_0 = 20 \text{ g/m}^2$. Define the dimensionless cutting surplus

$$\text{Da} \equiv \frac{k G \tau_{\text{dwell}}}{m_0} \approx 42 \quad (9)$$

Since $\text{Da} \gg 1$, the removal integral saturates: a pass strips essentially all of whatever it touches, and

$$\text{grit cut} = m_0 \times (\text{area swept}), \quad (10)$$

with technique entering only through whether $\text{Da} > 1$ at all. This is the formal statement of “coverage beats technique”. It also says the claim is *fragile in a specific place*: the worn lanes at the 18 in lead give $G = 0$ and hence $\text{Da} = 0$, the one regime where the argument inverts.

technique	G (Pa)	m_{cap} (g/m ²)	Da
baseline, base yield	4462	845	42.2
baseline, worn lane	1475	279	14.0
operator 12 in, base	3079	655	32.7
operator 12 in, worn	676	144	7.2
operator 18 in, base	101	33	1.7
operator 18 in, worn	0	0	0.0

The experimental data agree

Regressing the measured clean fraction on coverage across the four routing strategies gives

$$f_{\text{clean}} = 0.596 c + 0.025, \quad R^2 = 0.88, \quad (11)$$

over a range of c from 0.04 to 0.94. The saturated model predicts a proportionality; the slope of 0.60 rather than 1 is the delivery loss of Section 4—grit cut but not carried to the trough. The greedy “chase the dirtiest spot” strategy is the extreme case: it has the highest efficiency per unit work ($\eta = 0.858$) and the worst outcome (6% clean), because $c = 0.041$. Efficiency multiplies a quantity that is already saturated.

The time scale this implies

If the job is coverage-limited, the floor area $A = 4.2 \times 14 = 58.8 \text{ m}^2$ divided by the sweep rate $W_{\text{eff}} v$ is the whole time estimate:

$$T_{\text{min}} = \frac{A}{W_{\text{eff}} v} = \frac{58.8 \text{ m}^2}{0.22 \text{ m} \times 0.3 \text{ m/s}} = 891 \text{ s} \approx 15 \text{ min}, \quad (12)$$

which is both the episode length chosen in **SimConfig** and the time the operator reports. The agreement is a consistency check, not a validation—but it does locate the leverage: T_{min} is linear in $1/v$ and $1/W_{\text{eff}}$, and walking speed is the one factor in it that has never been measured.

4 Water sets transport, not cutting

Sustaining a film of depth h over the bay is a Manning normal-depth problem. With $n = 0.018$, cross slope $S = 0.015$, and a drainage width $W_d = 2L_x = 8.4 \text{ m}$,

$$Q = \frac{W_d}{n} \sqrt{S} h^{5/3} = 57.2 h^{5/3} \quad \Longleftrightarrow \quad h = \left(\frac{Q}{57.2} \right)^{3/5}. \quad (13)$$

The exponent $3/5$ is the whole story: water is a badly leveraged input.

supply	4 gpm	8 gpm	20 gpm
h (mm)	0.61	0.93	1.61
gravity bed shear ρghS (Pa)	0.09	0.14	0.24
transport length uh/v_s (m) at $u = 1$ m/s	0.20	0.31	0.54

Three consequences fall out, each matching a measured result.

1. A lone wand at 4 gpm sustains 0.6 mm and a bed shear of 0.09 Pa, *below* the 0.1 Pa deposit threshold. One operator cuts grit loose faster than the floor can carry it. Water is a shared resource, so operators should help each other superlinearly.
2. Raising supply $8 \rightarrow 20$ gpm ($2.5\times$) raises h by only $2.5^{3/5} = 1.73\times$ and leaves P —and therefore cutting—untouched. The experiment found exactly this: cutting identical at 92.3%, stuck grit identical at 0.088 kg, and delivery improving $79.8 \rightarrow 86.0\%$. The predicted delivery-deficit ratio $1/1.73 = 0.58$ brackets the observed $14.0/20.2 = 0.69$.
3. Since $L_t = uh/v_s \propto Q^{3/5}$, push distance scales as the $3/5$ power of standing water—the operator’s own “it depends on standing water,” with an exponent attached.

5 The two-stress structure, and why phases must not be separated

Adhered grit yields to *normal* stress at kilopascals; loose deposited grit yields to *tangential* stress at a Shields-type threshold of 0.1 Pa—four orders of magnitude apart. The jet supplies both, but in opposite proportions:

$$P_{\perp} \propto \cos^4 \theta d^{-2}, \quad \tau_{\parallel} \propto \sin \theta \|\text{momentum}\|. \quad (14)$$

Their ratio $\tau_{\parallel}/P_{\perp} \propto \tan \theta / \cos^3 \theta$ rises without bound as the wand lays over, which is the entire control trade-off.

Separating the job into a blast phase ($\theta = 12.6^\circ$) and a sweep phase ($\theta = 71.6^\circ$) fails for a reason visible in the deposition term. Grit in suspension settles at $v_s \|c\|$ and returns to the *deposited* layer, never the bound one; the surviving grit is therefore whatever has travelled less than $L_t \approx 0.3$ m before its film stalls. A combined pass re-entrains that grit within the same patch; a separated pass leaves it behind and must find it again later with a jet that, at 71.6° , delivers $P = 0.26$ kPa $= 0.10 Y_0$ and cannot cut anything it missed. Measured: $3.7\times$ more grit left stuck, 19 points dirtier.

6 Numerical guarantees

Two properties are load-bearing for the reward signal rather than cosmetic.

Conservation. All transport terms are in flux form with upwinding, so interior fluxes telescope and grit changes only through the trough. The reward is a mass difference; numerical diffusion would pay the agent for nothing.

Positivity. Explicit advection is CFL-limited, and the per-face velocity cap uses the *upwind* depth with a factor of two:

$$u_{\max} = \frac{\text{CFL} \cdot \Delta x}{2 \Delta t} = \frac{0.45 \times 0.07 \text{ m}}{2 \times (0.2 \text{ s}/16)} = 1.26 \text{ m/s}. \quad (15)$$

The factor of two is not conservatism: in 2D a cell can drain through four faces at once, so a per-face limit of CFL permits $4 \text{ CFL } h = 1.8 h$ of outflow per substep, and the positivity clamp would then manufacture water and sediment.

Inertia. The flow solver keeps the acceleration term (local inertial form, Bates et al. 2010) rather than using the diffusive-wave approximation. Dropping it forces water to relax instantly to its friction-balanced velocity, collapsing $L_t = uh/v_s$ from over a metre to centimetres the moment slurry leaves the impact patch. A scheme that cannot represent coasting cannot answer the question being asked.

7 Summary of the scaling laws

quantity	law	consequence
cutting pressure	$P \propto \cos^4 \theta d^{-2} e^{-d/\Lambda \cos \theta}$	angle dominates; cliff at 0.8 m
cutting onset	$G \approx \frac{1}{2} Y \varepsilon^2$	soft, C^1 threshold
effective swath	$W \propto L \sqrt{\ln(P/Y)}$	non-monotonic, peaks at 0.4 m
cutting surplus	$\text{Da} = kG\tau_d/m_0 \approx 42$	saturated: coverage is the objective
job time	$T = A/(W_{\text{eff}}v)$	linear in walk speed
film depth	$h \propto Q^{3/5}$	water is badly leveraged
transport length	$L_t = uh/v_s \propto Q^{3/5}$	extra water buys delivery, not cutting
stress ratio	$\tau_{\parallel}/P_{\perp} \propto \tan \theta \sec^3 \theta$	cut and push cannot be maximised together

The one number still missing. Every result above is a ranking, and rankings are robust. The single quantity that converts them into a clock is the walking speed v , which enters $T_{\min}$ linearly and τ_{dwell} inversely—and it is the one term in the model that has never been measured.